

LAB EXPERIMENT RECORD

SET 4: PRODUCT INVENTORY MANAGEMENT

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date / Day	08 October 2026 / Thursday

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Product Category	Product Price
1	Product A	250	Electronics	500
2	Product B	175	Grocery	150
3	Product C	300	Electronics	750
4	Product D	200	Grocery	250
5	Product E	220	Home	400

Note: The original question supplies Product ID, Product Name, and Quantity Available. Product Category and Product Price were added as sample values only because Programs 2 and 3 require those fields.

PROGRAM 1: Bar Chart of Product Inventory

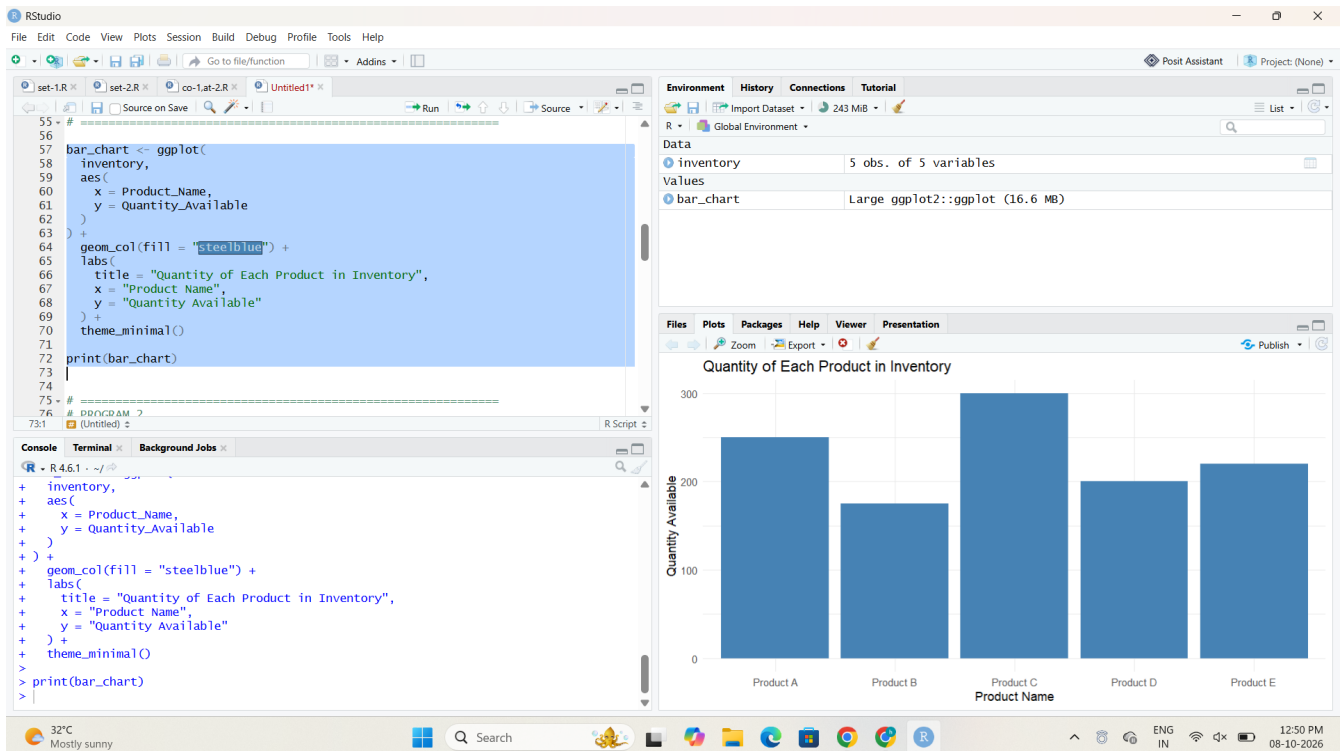
Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Product Inventory Data
inventory <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C",
                  "Product D", "Product E"),
  Quantity_Available = c(250, 175, 300, 200, 220)
)

library(ggplot2)

ggplot(inventory, aes(x = Product_Name,
                     y = Quantity_Available)) +
  geom_col(fill = "steelblue") +
  labs(title = "Quantity of Each Product in Inventory",
       x = "Product Name",
       y = "Quantity Available") +
  theme_minimal()
```

Output 1: Quantity of Each Product in Inventory



Observation: Product C has the highest available quantity at 300 units, while Product B has the lowest at 175 units.

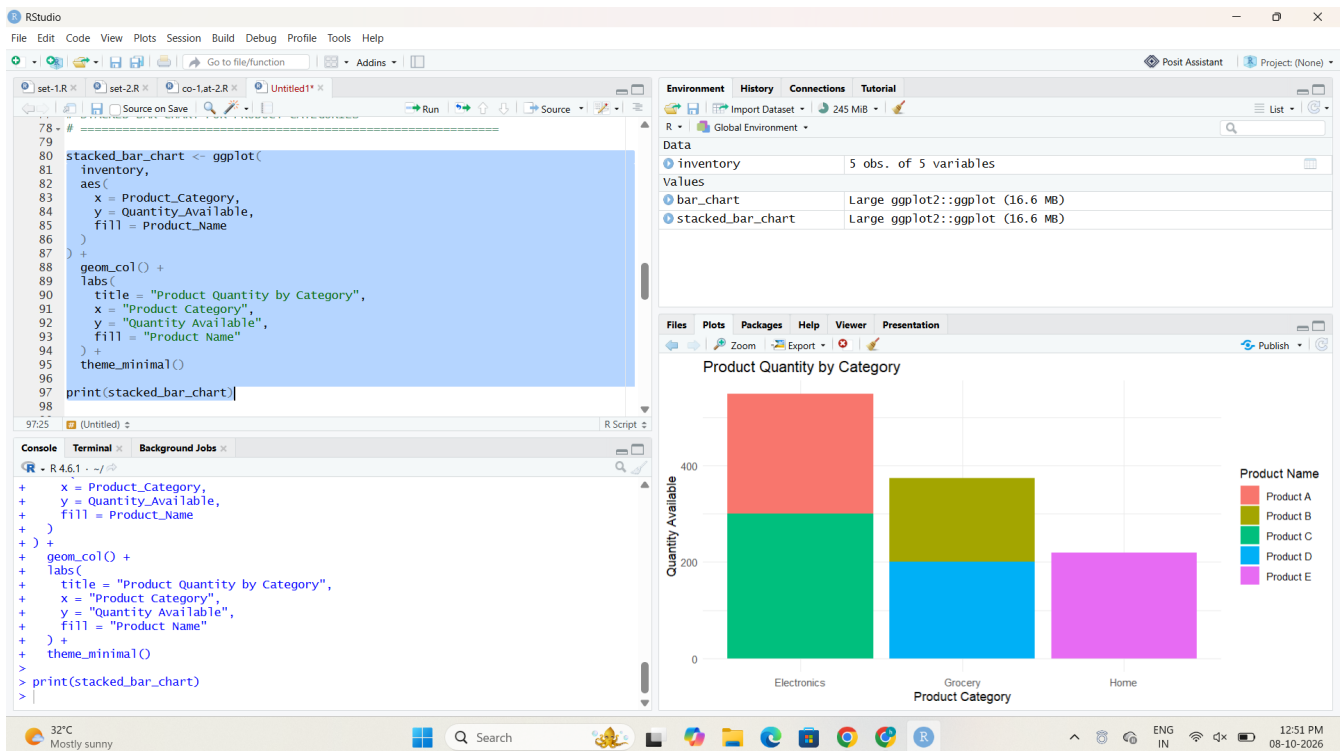
PROGRAM 2: Stacked Bar Chart by Product Category

Generate a stacked bar chart to show the quantity of each product within different product categories.

```
# Stacked Bar Chart
stacked_bar_chart <- ggplot(
  inventory,
  aes(x = Product_Category,
      y = Quantity_Available,
      fill = Product_Name)
) +
  geom_col() +
  labs(title = "Product Quantity by Category",
       x = "Product Category",
       y = "Quantity Available",
       fill = "Product Name") +
  theme_minimal()

print(stacked_bar_chart)
```

Output 2: Product Quantity by Category



Observation: Electronics contains Product A and Product C, Grocery contains Product B and Product D, and Home contains Product E. The stacked bars show each product's contribution to its category total.

PROGRAM 3: Product Price vs Quantity Available

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

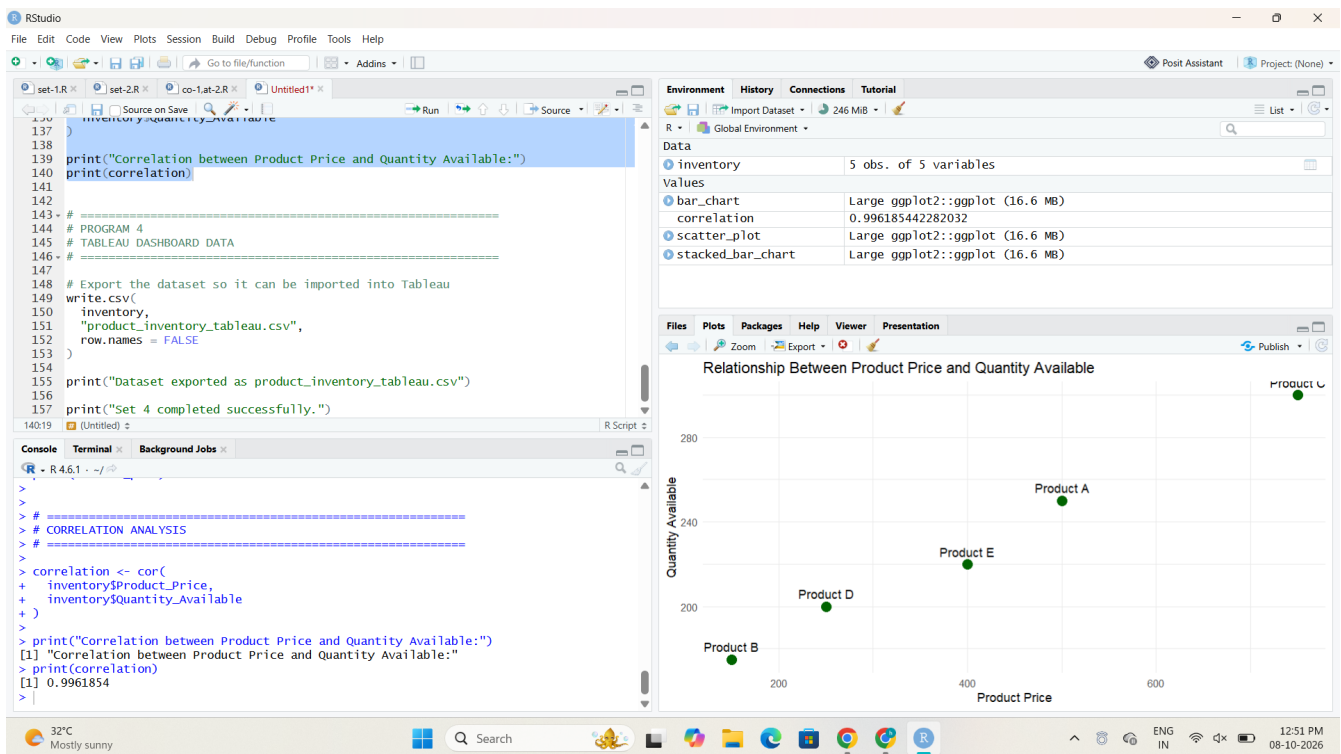
```
# Scatter Plot
scatter_plot <- ggplot(
  inventory,
  aes(x = Product_Price,
      y = Quantity_Available,
      label = Product_Name)
) +
  geom_point(size = 4, color = "darkgreen") +
  geom_text(vjust = -0.8) +
  labs(
    title = "Relationship Between Product Price and Quantity Available",
    x = "Product Price",
    y = "Quantity Available"
  ) +
  theme_minimal()

print(scatter_plot)

# Correlation
correlation <- cor(
  inventory$Product_Price,
  inventory$Quantity_Available
)

print(correlation)
```

Output 3: Relationship Between Product Price and Quantity Available



Finding: The calculated correlation is approximately 0.9962, indicating a very strong positive relationship between the sample product prices and quantities. This conclusion applies only to the sample price values added for the exercise.

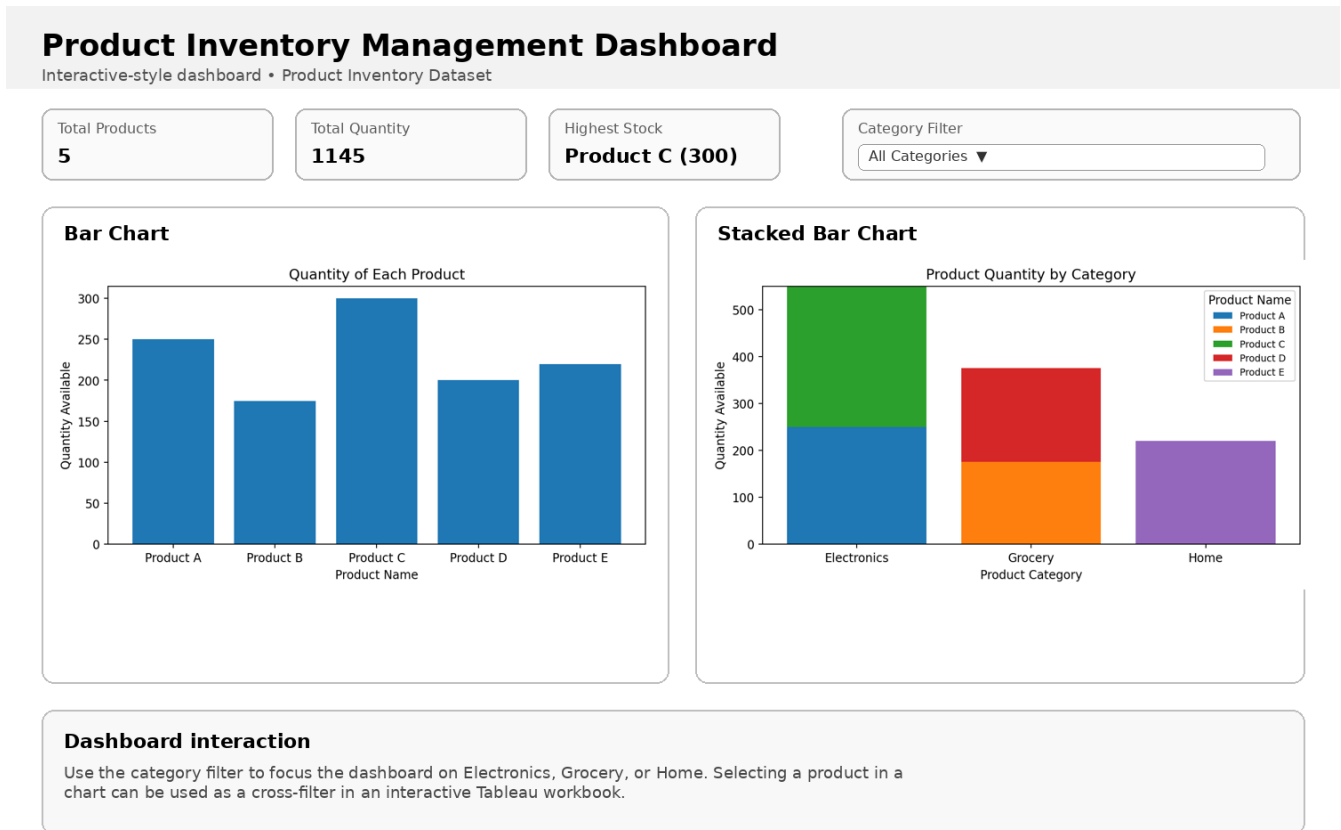
PROGRAM 4: Product Inventory Tableau Dashboard

Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

```
# Export the dataset for Tableau
write.csv(
  inventory,
  "product_inventory_tableau.csv",
  row.names = FALSE
)

print("Dataset exported as product_inventory_tableau.csv")
```

Output 4: Product Inventory Tableau Dashboard



Observation: The dashboard combines the product inventory bar chart and category-wise stacked bar chart in one interactive-style view. A category filter can be used to focus the analysis on a selected product category. The dashboard image is an illustrative documentation image based on the same dataset.

SET 4 PROCEDURE

1. Create the Product Inventory dataset in R.
2. Use ggplot2 to generate the bar chart, stacked bar chart, and scatter plot.
3. Run each program separately and verify the generated outputs.
4. Export the inventory dataset as a CSV file for Tableau.
5. Create a dashboard combining the bar chart and stacked bar chart and provide interactive filtering.
6. Capture the outputs and insert them in the experiment record.

SET 4 RESULT

The Product Inventory data was successfully visualized using a bar chart, stacked bar chart, and scatter plot. The relationship between product price and quantity was analyzed using correlation, and a dashboard-style Tableau representation was prepared for interactive exploration.